US 20250274676A1

(19) United States

(12) Patent Application Publication

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(10) Pub. No.: US 2025/0274676 A1

(43) Pub. Date: Aug. 28, 2025

(54) IMAGE CAPTURING SYSTEM CAPABLE OF PERFORMING IMAGE CAPTURING IN DESIRED CHARGE ACCUMULATION TIME IN IMAGE SENSOR WHILE SUPPRESSING GENERATION OF STRIPES, IMAGE CAPTURING APPARATUS, LIGHTING DEVICE, METHOD OF CONTROLLING IMAGE CAPTURING SYSTEM, AND STORAGE MEDIUM

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(21) Appl. No.: **19/049,170**

(22) Filed: **Feb. 10, 2025**

(30) **Foreign Application Priority Data**

Feb. 22, 2024 (JP) 2024-025507

Publication Classification

(51) **Int. Cl.**
H04N 25/50 (2023.01)
H04N 25/76 (2023.01)

(52) **U.S. Cl.**
CPC *H04N 25/50* (2023.01); *H04N 25/76* (2023.01)

(57) **ABSTRACT**

An image capturing system including an image capturing apparatus and a lighting device, which are communicably connected to each other. The image capturing apparatus includes an image sensor and a system controller configured to control a charge accumulation time in the image sensor. The lighting device includes a light emission section and a video light controller configured to control light emission from the light emission section by performing PWM control. The system controller determines a frequency in the PWM control based on the charge accumulation time and notifies the determined frequency to the video light controller. The video light controller controls light emission from the light emission section at the frequency in the PWM control, which is notified from the system controller.

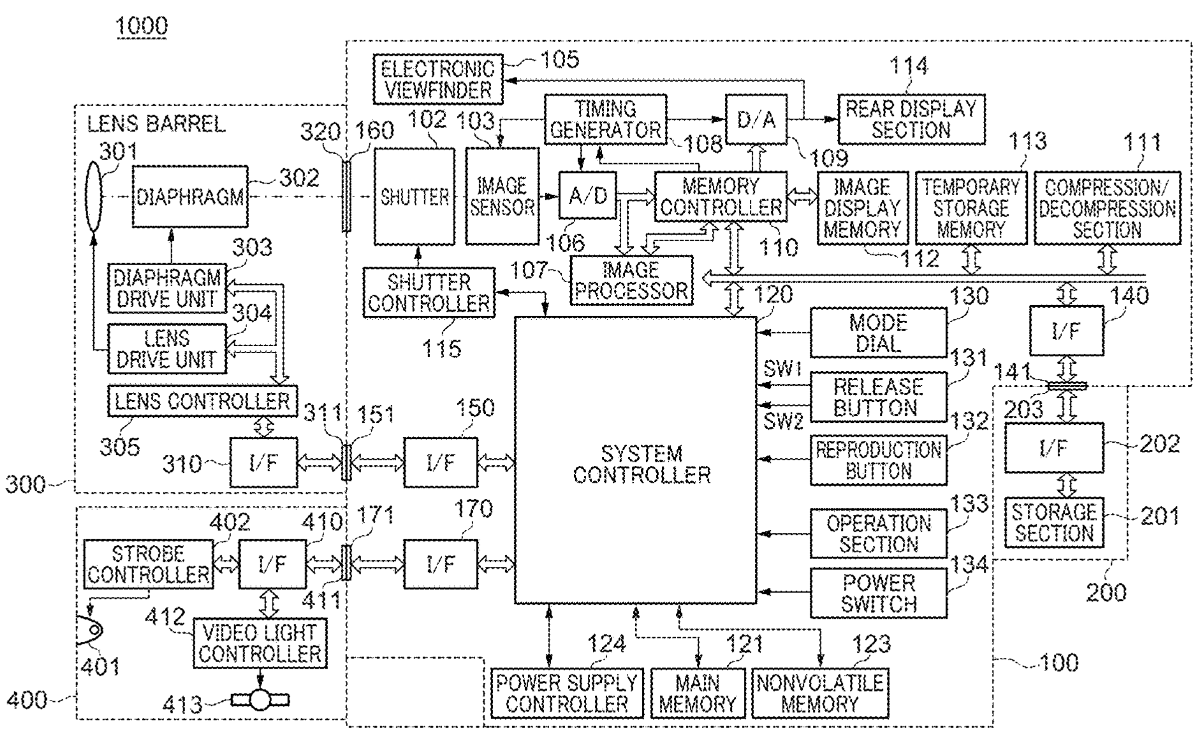

FIG. 1

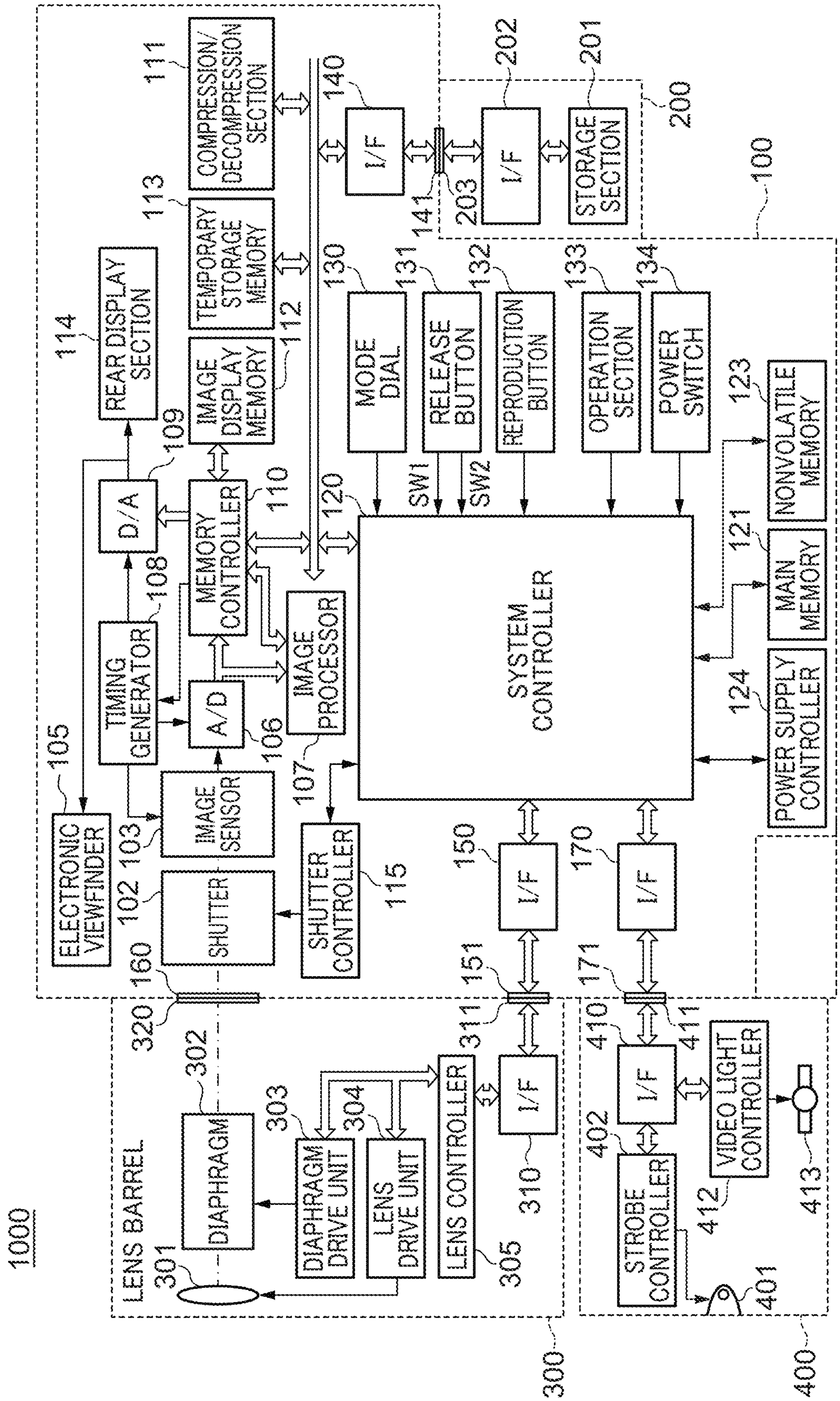
1000
LENS BARREL
301
DIAPHRAGM
302
DIAPHRAGM DRIVE UNIT
303
LENS DRIVE UNIT
304
LENS CONTROLLER
305
I/F
310
311
320
300
STROBE CONTROLLER
402
401
I/F
410
411
VIDEO LIGHT CONTROLLER
412
413
400
ELECTRONIC VIEWFINDER
105
SHUTTER
102
IMAGE SENSOR
103
SHUTTER CONTROLLER
115
TIMING GENERATOR
108
A/D
106
IMAGE PROCESSOR
107
D/A
109
MEMORY CONTROLLER
110
REAR DISPLAY SECTION
114
IMAGE DISPLAY MEMORY
112
TEMPORARY STORAGE MEMORY
113
COMPRESSION/ DECOMPRESSION SECTION
111
I/F
140
141
203
I/F
202
STORAGE SECTION
201
200
MODE DIAL
130
RELEASE BUTTON
131
SW1
SW2
REPRODUCTION BUTTON
132
OPERATION SECTION
133
POWER SWITCH
134
SYSTEM CONTROLLER
120
NONVOLATILE MEMORY
123
MAIN MEMORY
121
POWER SUPPLY CONTROLLER
124
I/F
150
151
I/F
170
171
160
100

FIG. 2

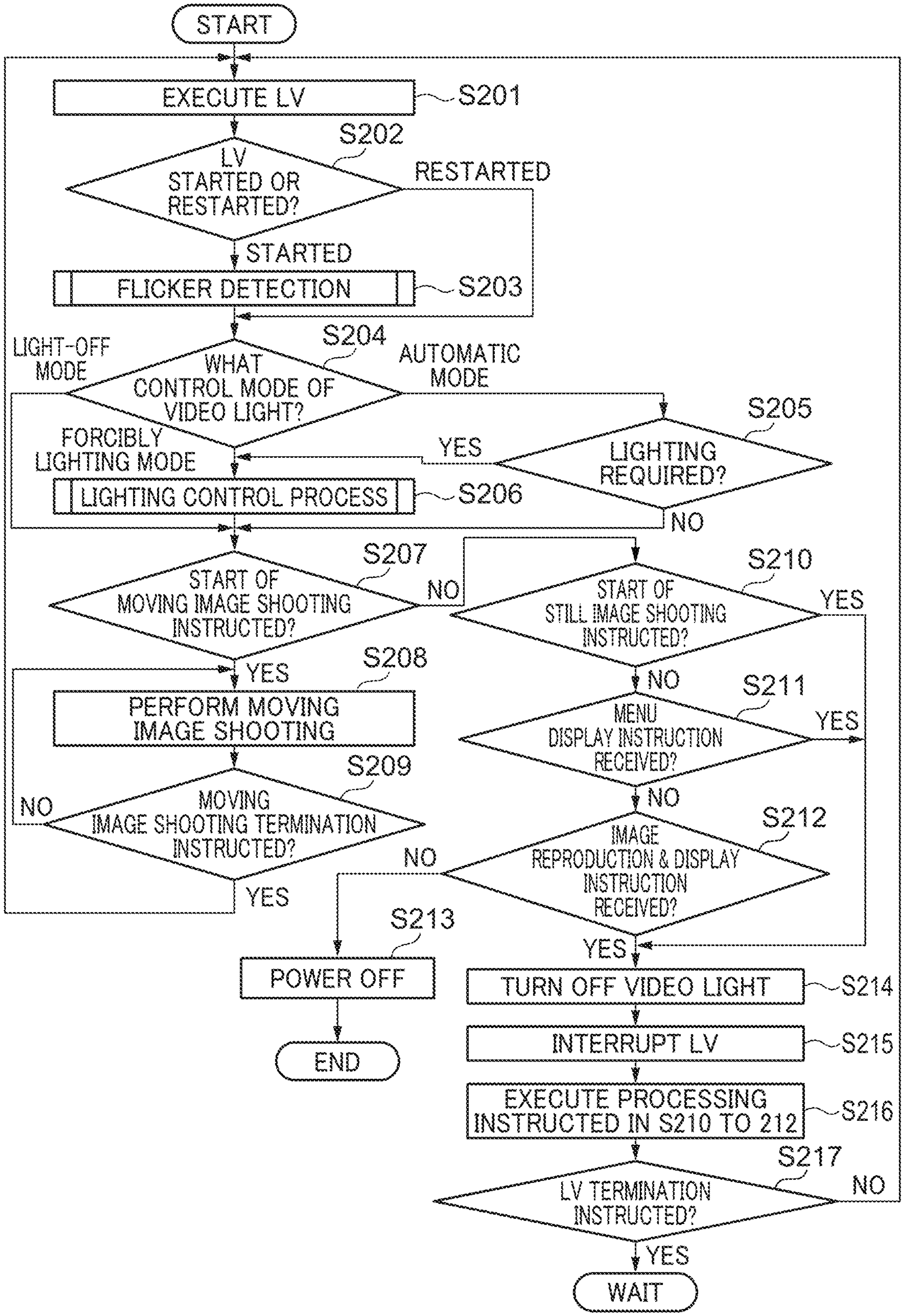
START
EXECUTE LV
S201
S202
LV STARTED OR RESTARTED?
RESTARTED
STARTED
FLICKER DETECTION
S203
S204
LIGHT-OFF MODE
WHAT CONTROL MODE OF VIDEO LIGHT?
AUTOMATIC MODE
S205
FORCIBLY LIGHTING MODE
YES
LIGHTING REQUIRED?
LIGHTING CONTROL PROCESS
S206
NO
S207
START OF MOVING IMAGE SHOOTING INSTRUCTED?
NO
S210
START OF STILL IMAGE SHOOTING INSTRUCTED?
YES
S208
YES
PERFORM MOVING IMAGE SHOOTING
NO
S211
MENU DISPLAY INSTRUCTION RECEIVED?
YES
S209
NO
MOVING IMAGE SHOOTING TERMINATION INSTRUCTED?
NO
S212
NO
IMAGE REPRODUCTION & DISPLAY INSTRUCTION RECEIVED?
YES
S213
YES
POWER OFF
TURN OFF VIDEO LIGHT
S214
END
INTERRUPT LV
S215
EXECUTE PROCESSING INSTRUCTED IN S210 TO 212
S216
S217
LV TERMINATION INSTRUCTED?
NO
YES
WAIT

FIG. 3A

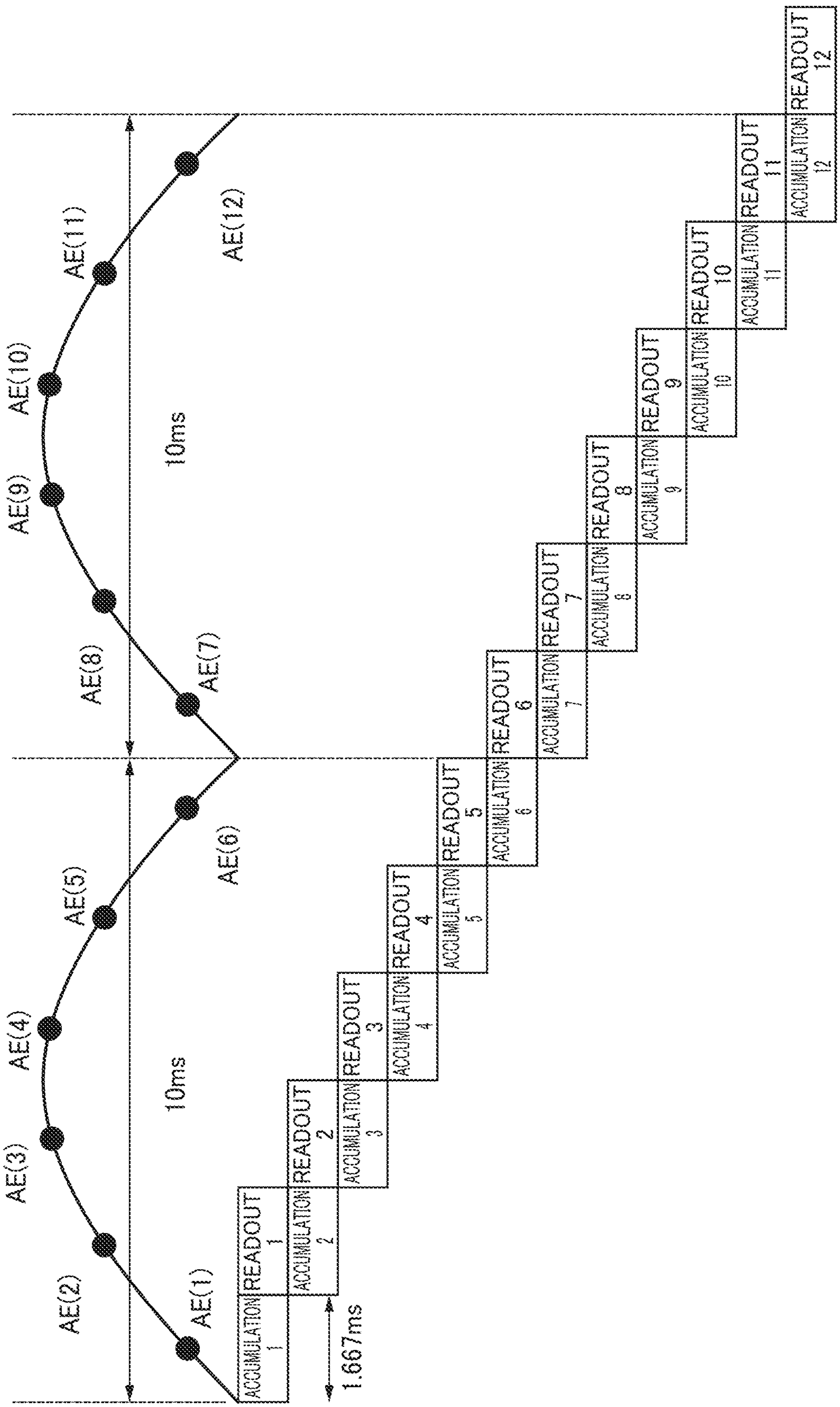
AE(1)
AE(2)
AE(3)
AE(4)
AE(5)
AE(6)
AE(7)
AE(8)
AE(9)
AE(10)
AE(11)
AE(12)
10ms
10ms
1.667ms
ACCUMULATION 1
READOUT 1
ACCUMULATION 2
READOUT 2
ACCUMULATION 3
READOUT 3
ACCUMULATION 4
READOUT 4
ACCUMULATION 5
READOUT 5
ACCUMULATION 6
READOUT 6
ACCUMULATION 7
READOUT 7
ACCUMULATION 8
READOUT 8
ACCUMULATION 9
READOUT 9
ACCUMULATION 10
READOUT 10
ACCUMULATION 11
READOUT 11
ACCUMULATION 12
READOUT 12

FIG. 3B

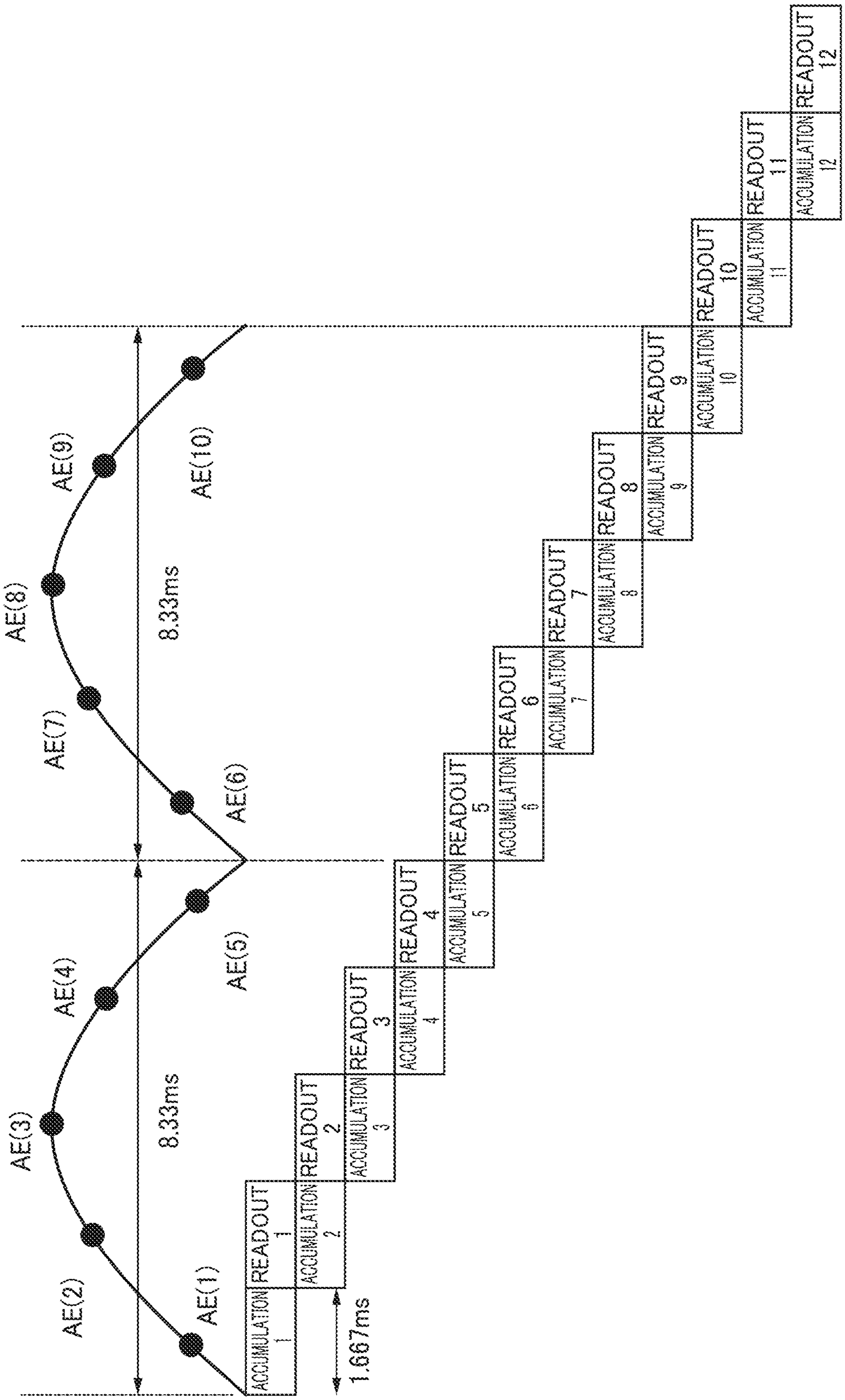
AE(1)
AE(2)
AE(3)
AE(4)
AE(5)
AE(6)
AE(7)
AE(8)
AE(9)
AE(10)
8.33ms
8.33ms
1.667ms
ACCUMULATION 1
READOUT 1
ACCUMULATION 2
READOUT 2
ACCUMULATION 3
READOUT 3
ACCUMULATION 4
READOUT 4
ACCUMULATION 5
READOUT 5
ACCUMULATION 6
READOUT 6
ACCUMULATION 7
READOUT 7
ACCUMULATION 8
READOUT 8
ACCUMULATION 9
READOUT 9
ACCUMULATION 10
READOUT 10
ACCUMULATION 11
READOUT 11
ACCUMULATION 12
READOUT 12

FIG. 4A

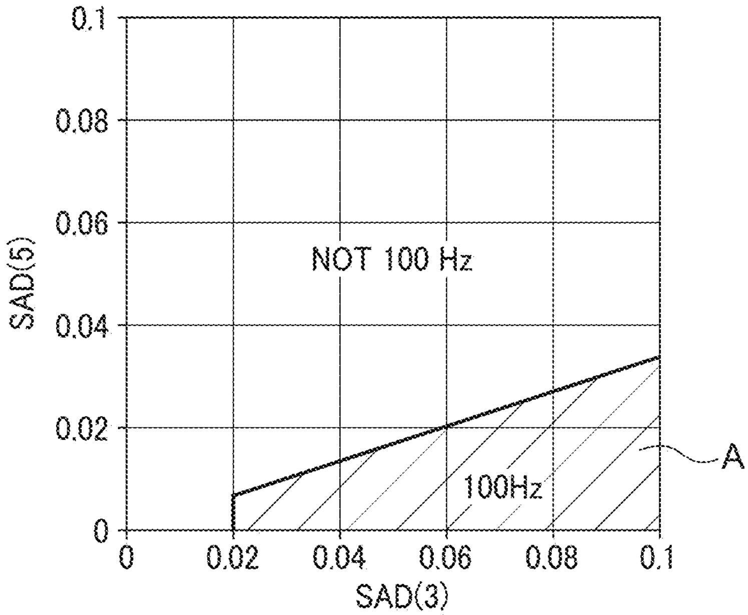
0.1
0.08
0.06
0.04
0.02
0
SAD(5)
NOT 100 Hz
100Hz
A
0
0.02
0.04
0.06
0.08
0.1
SAD(3)

FIG. 4B

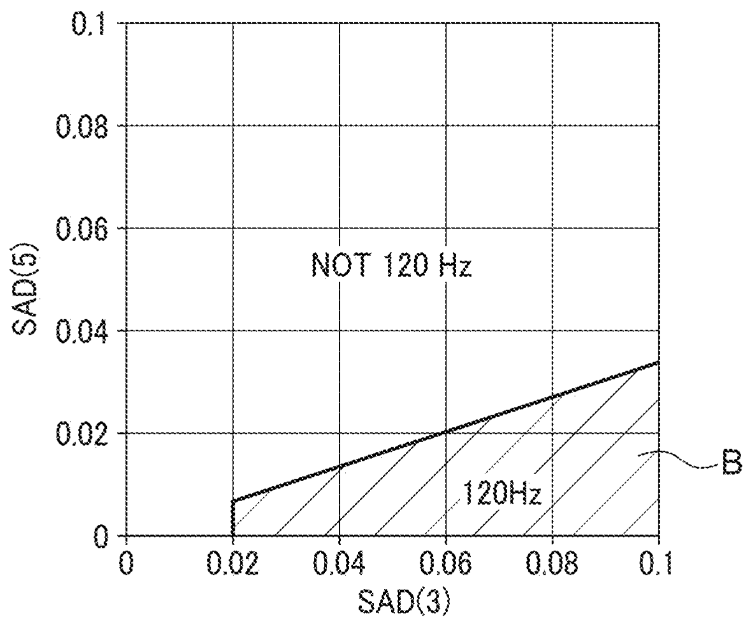
0.1
0.08
0.06
0.04
0.02
0
SAD(5)
NOT 120 Hz
120Hz
B
0
0.02
0.04
0.06
0.08
0.1
SAD(3)

FIG. 4C

	100Hz	NOT 100 Hz
120Hz	DC	120Hz
NOT 120 Hz	100Hz	DC

FIG. 5

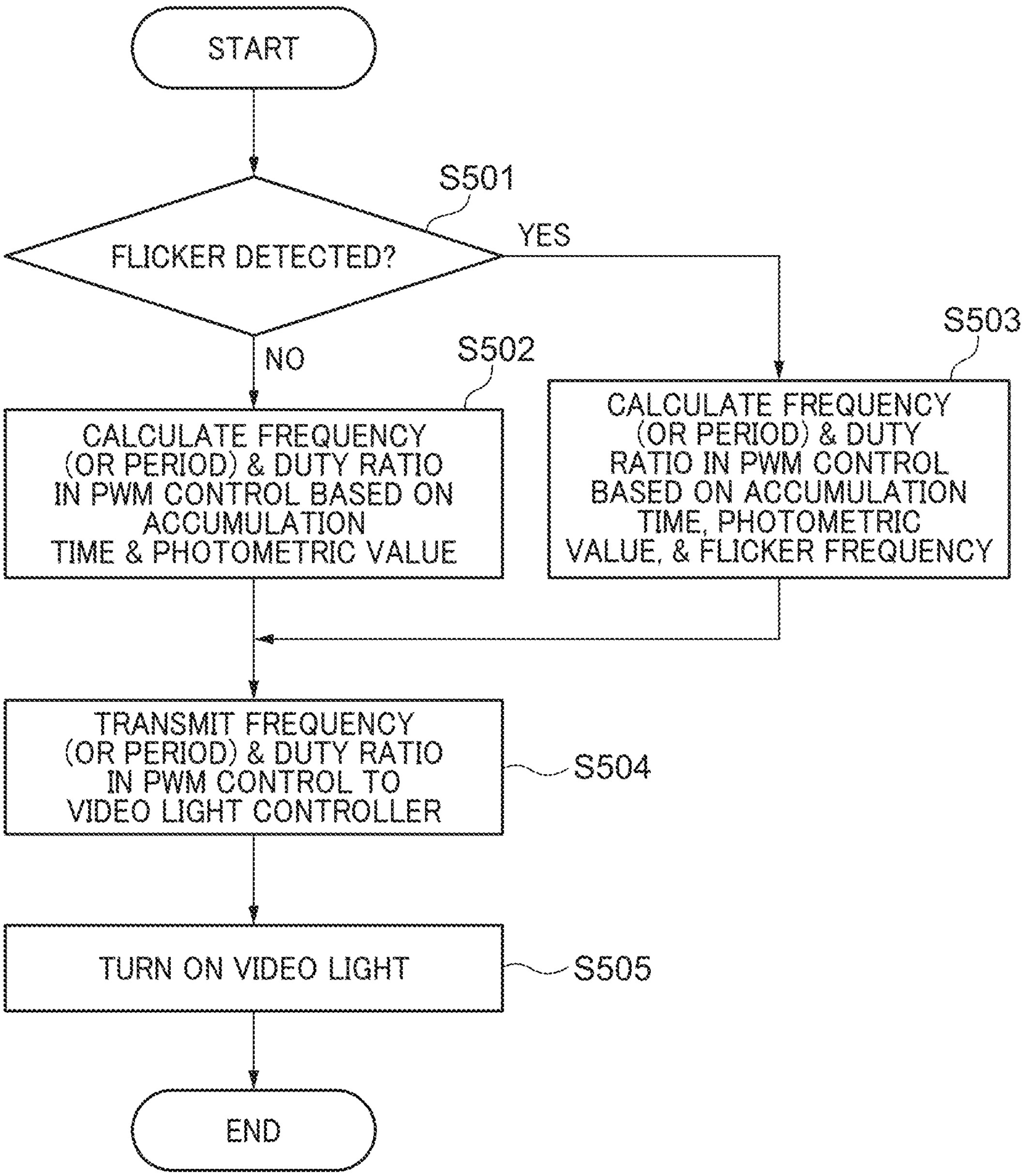
START
FLICKER DETECTED?
S501
YES
NO
S502
CALCULATE FREQUENCY (OR PERIOD) & DUTY RATIO IN PWM CONTROL BASED ON ACCUMULATION TIME & PHOTOMETRIC VALUE
S503
CALCULATE FREQUENCY (OR PERIOD) & DUTY RATIO IN PWM CONTROL BASED ON ACCUMULATION TIME, PHOTOMETRIC VALUE, & FLICKER FREQUENCY
TRANSMIT FREQUENCY (OR PERIOD) & DUTY RATIO IN PWM CONTROL TO VIDEO LIGHT CONTROLLER
S504
TURN ON VIDEO LIGHT
S505
END

IMAGE CAPTURING SYSTEM CAPABLE OF PERFORMING IMAGE CAPTURING IN DESIRED CHARGE ACCUMULATION TIME IN IMAGE SENSOR WHILE SUPPRESSING GENERATION OF STRIPES, IMAGE CAPTURING APPARATUS, LIGHTING DEVICE, METHOD OF CONTROLLING IMAGE CAPTURING SYSTEM, AND STORAGE MEDIUM

BACKGROUND

Technical Field

[0001] The aspect of the embodiments relates to, in an image capturing system including a lighting device and an image capturing apparatus, a technique for controlling light emission from the lighting device.

Description of the Related Art

[0002] In a dark place, shooting is performed using a lighting device that illuminates an object and its background. In doing this, as the lighting device used, a strobe device is used in the case of still image shooting, a video light is used in the case of moving image shooting, and these are used in a state incorporated in an image capturing apparatus (camera), or in a state physically or wirelessly connected to the image capturing apparatus.

[0003] Here, as the video light, one using an incandescent light as the light source has been conventionally used, but in recent years, one using a light emitting diode (LED) as the light source has been increasingly used. By using the LED for the light source, it is possible to easily adjust the intensity of light irradiated on an object by changing a voltage to be applied, and further by performing pulse width modulation (PWM) control.

[0004] In the PWM control, the amount of light emission is adjusted by repeating on/off of the voltage to be input to the LED to thereby continuously perform lighting/non-lighting of the LED. Therefore, it is known that, in a camera using a CMOS sensor which acquires image signals by using the rolling shutter method, stripe-shaped brightness differences similar to flicker appear in a captured image due to light irradiated from the video light to an object, depending on an image capturing condition.

[0005] To solve the problem, as a method of reducing the stripes generated on an object image due to irradiation of light from the video light, application of the conventional countermeasure against flicker is considered. For example, Japanese Laid-Open Patent Publication (Kokai) No. 2020-10317 discloses a technique for detecting flicker based on a plurality of images obtained by driving an image sensor at a predetermined period and setting a charge accumulation time (shutter speed) in the image sensor to a value at which an influence of the detected flicker is reduced. By using this technique, it is considered that, by controlling the charge accumulation time in the image sensor based on a frequency (period) at a time when the video light is caused to emit light according to the PWM control, it is possible to perform live view display in which generation of stripes is suppressed.

[0006] However, in the above-mentioned conventional technique, the charge accumulation time in the image sensor is limited to the frequency (period) in the PWM control for causing the video light to emit light, and hence for example, a situation can be caused where it is impossible to perform moving image shooting in a desired charge accumulation time.

SUMMARY

[0007] According to a first aspect of the embodiments, there is provided an image capturing system including an image capturing apparatus and a lighting device, which are communicably connected to each other, wherein the image capturing apparatus includes an image sensor, and at least one memory configured to store instructions, and at least one processor in communication with the at least one memory and configured to execute the instructions to function as: a first control unit configured to control a charge accumulation time in the image sensor, wherein the lighting device includes a light emission section; and at least one memory configured to store instructions, and at least one processor in communication with the at least one memory and configured to execute the instructions to function as: a second control unit configured to control light emission from the light emission section by performing PWM control, wherein the first control unit determines a frequency in the PWM control, based on the charge accumulation time, and notifies the determined frequency to the second control unit, and wherein the second control unit controls light emission from the light emission section at the frequency in the PWM control, which is notified from the first control unit.

[0008] According to a second aspect of the embodiments, there is provided an image capturing system including an image capturing apparatus and a lighting device, which are communicably connected to each other, wherein the image capturing apparatus includes an image sensor, and at least one memory configured to store instructions, and at least one processor in communication with the at least one memory and configured to execute the instructions to function as: a first control unit configured to control a charge accumulation time in the image sensor, wherein the lighting device includes a light emission section, and at least one memory configured to store instructions, and at least one processor in communication with the at least one memory and configured to execute the instructions to function as: a second control unit configured to control light emission from the light emission section by performing PWM control, wherein the first control unit notifies the charge accumulation time to the second control unit, and wherein the second control unit determines a frequency in the PWM control, based on the charge accumulation time, and controls light emission from the light emission section.

[0009] Further features of the disclosure will become apparent from the following description of exemplary embodiments (with reference to the attached drawings).

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. **1** is a block diagram showing a schematic configuration of an image capturing system according to an embodiment.

[0011] FIG. **2** is a flowchart of a shooting process performed by the image capturing system.

[0012] FIGS. **3**A and **3**B are diagrams useful in explaining a relationship between charge accumulation in an image sensor and readout therefrom, and photometric values.

[0013] FIGS. **4**A to **4**C are maps for determining a flicker environment and a determination table.

[0014] FIG. **5** is a flowchart of a lighting control process for a video light.

DESCRIPTION OF THE EMBODIMENTS

[0015] Hereinafter, embodiments will be described in detail with reference to the attached drawings. Note, the following embodiments are not intended to limit the scope of the disclosure. Multiple features are described in the embodiments, but limitation is not made to a disclosure that requires all such features, and multiple such features may be combined as appropriate. Furthermore, in the attached drawings, the same reference numerals are given to the same or similar configurations, and redundant description thereof is omitted. Here, as an image capturing system according to the disclosure, a configuration in which a lens barrel (interchangeable lens) and a lighting device are attached to an image capturing apparatus, such as a digital camera, will be described.

[0016] Here, in the following description, the image capturing apparatus refers to a main body part (apparatus body) of the image capturing apparatus equipped with an image sensor. However, it is assumed for convenience that the image capturing apparatus also refers to an image capturing apparatus, such as a compact digital camera or a digital video camera, which is formed by integrating a photographic lens with the apparatus body.

[0017] FIG. **1** is a block diagram showing a schematic configuration of an image capturing system **1000** according to an embodiment. The image capturing system **1000** is formed by an image capturing apparatus **100**, a lens barrel **300** and a lighting device **400**, which are attached to the image capturing apparatus **100**, and an external storage medium **200** accommodated in the image capturing apparatus **100**.

[0018] The image capturing apparatus **100** is, specifically, a digital camera capable of shooting a moving image, and further, here, a mirrorless-type digital single-lens reflex camera is assumed. The image capturing apparatus **100** includes a lens mount **160**, an image sensor **103**, a shutter **102**, a shutter controller **115**, an A/D converter **106**, an image processor **107**, a memory controller **110**, a D/A converter **109**, an image display memory **112**, and a temporary storage memory **113**. The image capturing apparatus **100** further includes a compression/decompression section **111**, a timing generator **108**, an electronic viewfinder **105**, a rear display section **114**, a system controller **120**, a main memory **121**, a nonvolatile memory **123**, an operation section **133**, a mode dial **130**, and a release button **131**. The image capturing apparatus **100** also includes a reproduction button **132**, a power switch **134**, a power supply controller **124**, a first camera interface (I/F) **140**, a second camera I/F **150**, a third camera I/F **170**, a first camera connector **141**, a second camera connector **151**, and a third camera connector **171**.

[0019] The external storage medium **200** includes a media connector **203**, a media I/F **202**, and a storage section **201**.

[0020] The lens barrel **300** includes a camera mount **320**, a lens connector **311**, a lens I/F **310**, a photographic lens group **301**, a diaphragm **302**, a diaphragm drive unit **303**, a lens drive unit **304**, and a lens controller **305**.

[0021] The lighting device **400** includes a strobe connector **411**, a strobe I/F **410**, a strobe controller **402**, a strobe light emission section **401**, a video light controller **412**, and a video light **413**.

[0022] In the image capturing apparatus **100**, the image sensor **103** is a charge accumulation-type photoelectric conversion device, such as a CMOS. An object optical image formed on an imaging surface of the image sensor **103**, which has been incident through the lens barrel **300** is converted to analog image signals by photoelectric conversion, and the generated analog signals are output to the A/D converter **106**.

[0023] The shutter **102** includes a front curtain and a rear curtain as mechanical components, and an actuator which drives the front curtain and the rear curtain, and opens and blocks an optical path between the lens barrel **300** and the image sensor **103** by driving the front curtain and the rear curtain. Note that the shutter **102** is not limited to the mechanical configuration that drives the front curtain and the rear curtain, but can be realized by a function of resetting or reading out image data captured by the image sensor **103** according to electrical control. The shutter controller **115** controls driving of the shutter **102** (the front curtain and the rear curtain) according to a command from the system controller **120**.

[0024] The A/D converter **106** converts the analog image signals output from the image sensor **103** to digital image data and outputs the generated digital image data to the image processor **107** and the memory controller **110**. The image processor **107** performs various kinds of processing operations, such as white balance adjustment processing and gradation processing, on the digital image data sent from the A/D converter **106**. The memory controller **110** controls writing and reading of a variety of data acquired from the A/D converter **106**, the image processor **107**, or the compression/decompression section **111** into and from the image display memory **112** or the temporary storage memory **113**.

[0025] The image display memory **112** temporarily stores display digital image data of an image to be displayed on the electronic viewfinder **105** and the rear display section **114**. The display digital image data written in the image display memory **112** is sent to the D/A converter **109** via the memory controller **110**. The D/A converter **109** converts the display digital image data from digital to analog to generate display analog image data and supplies the generated display analog image data to the electronic viewfinder **105** and the rear display section **114**.

[0026] The temporary storage memory **113** temporarily stores image data output from the image processor **107**, image data read out from the external storage medium **200**, and the like, and is also used as a work area for the system controller **120**. The compression/decompression section **111** loads image data stored in the temporary storage memory **113** and compresses or decompresses the image data by using predetermined image compression method and image decompression method, in a state adapted to a various kinds of uses.

[0027] The timing generator **108** generates operation signals (control signals, such as a clock signal) to be supplied to the image sensor **103**, the A/D converter **106**, the D/A converter **109**, and so forth. Further, the timing generator **108** controls accumulation of electric charges in the image sensor **103** by controlling timing of resetting the charges accumulated in the image sensor **103**. The electronic viewfinder **105** and the rear display section **114** are each implemented e.g. by a liquid crystal panel or an organic EL panel and display a menu screen and so forth for performing a variety of settings for an object image and the image capturing apparatus **100**. The electronic viewfinder **105** employs a configuration that enables a user to confirm displayed contents by bringing an eye into contact with the electronic viewfinder **105**, and the rear display section **114** is arranged on a rear surface of the image capturing apparatus **100** via a variangle mechanism and a tilt mechanism.

[0028] The system controller **120** controls the overall operation of not only the image capturing apparatus **100**, but also the image capturing system **1000**. The specific example of processing operations performed by the system controller **120** will be described hereafter, but the processing operations performed by the system controller **120** are not limited to the following example.

[0029] The system controller **120** performs photometric calculation using image data acquired by the image sensor **103**, calculates a luminance value of an object (obtains luminance information), and determines exposure conditions of the image sensor **103**. As the parameters for determining the exposure conditions, an aperture value, a shutter speed, a photographing sensitivity (ISO sensitivity), and so forth are used. Thus, proper exposure for the image sensor **103** is controlled. Further, the system controller **120** performs, based on a result of focus detection on an object, position control of the lens group of the lens barrel **300** to focus on the object.

[0030] The system controller **120** controls the charge accumulation time in the image sensor **103** (shutter speed) by particularly controlling the shutter **102** when shooting a still image, and by particularly controlling the image sensor **103** when acquiring a live view image and shooting a moving image. The system controller **120** instructs the strobe controller **402** and the video light controller **412** based on a photometric calculation result, the exposure conditions, and a state of the lighting device **400** to thereby control a light emission amount and light emission timing of the strobe light emission section **401** and the video light **413**.

[0031] The system controller **120** controls the start and end of moving image recording according to an input signal from the operation section **133** and further can perform shooting using outputs from pixels on the whole imaging surface of the image sensor or shooting using part of the outputs, which are cut out from the pixels, by switching the image capturing range by the image sensor **103**. The system controller **120** performs, in still image shooting and moving image shooting, control to display the live view image for confirming an object, on the electronic viewfinder **105** or the rear display section **114**. Further, the system controller **120** detects presence/absence and frequency of flicker in a shooting environment by using image data obtained by the image sensor **103**.

[0032] The other component elements of the image capturing apparatus **100** will be described. The main memory **121** is e.g. a read only memory (ROM) and stores information concerning proper exposure with respect to a luminance value (table data and a program diagram) and data associated with the operations of the image capturing apparatus **100**, such as constants for the operations executed by the image capturing apparatus **100**, a variety of exposure conditions, and arithmetic expression. The nonvolatile memory **123** is e.g. an Electrically Erasable Programmable Read-Only Memory (EEPROM) and stores a variety of settings of the image capturing system **1000**.

[0033] The operation section **133** is formed by buttons, switches, a dial, a touch panel, for inputting various kinds of operation instructions to the system controller **120**, a sight line detection device, and an audio recognition device, or a combination of these. Note that the operation members included in the operation section **133** are operation members other than the mode dial **130**, the release button **131**, the reproduction button **132**, and the power switch **134**, separately illustrated in FIG. **1**.

[0034] The mode dial **130** is a rotary-type operation member used when a desired shooting mode is set from within a plurality of shooting modes which can be set for the image capturing apparatus **100**. In the image capturing apparatus **100**, it is possible to set a still image mode for shooting a still image and a moving image mode for recording a moving image. Further, in the shooting modes of still images and moving images, it is possible to set a variety of modes in which the exposure parameters can be automatically or manually set, such as a full-auto mode, a program mode, an aperture priority mode, a shutter speed priority mode, and a manual mode, and necessity/unnecessity of strobe light emission in still image shooting.

[0035] The release button **131** is an operation member for instructing the start of a preparation operation and the start of actual shooting in still image shooting and moving image shooting. The release button **131** generates a SW**1** signal when a first stroke (half-pressing) thereof is performed, and when the SW**1** signal is detected, the system controller **120** starts the shooting preparation operation. In the shooting preparation operation, focus control, exposure control, auto white balance (AWB) processing, and so forth are performed, and when moving image shooting is performed, lighting control and light emission control for the video light **413** of the lighting device **400** are performed as the exposure control, as required. The release button **131** generates a SW**2** signal when a second stroke (fully pressing) thereof is performed. When the SW**2** signal is detected, the system controller **120** performs a series of processing, ranging from the exposure processing on the image sensor **103** to the processing for storing image data in the external storage medium **200**.

[0036] The reproduction button **132** is an operation member that instructs to start reproduction processing for reading image data from the temporary storage memory **113** or the external storage medium **200** and displaying the read image data on the rear display section **114**. The power switch **134** is an operation member that switches on/off of supply of electric power from a power supply section (not shown), such as a battery, to the components of the image capturing apparatus **100**. Note that when the power switch **134** is turned on, electric power can be supplied not only to the image capturing apparatus **100**, but also to the lens barrel **300**, the external storage medium **200**, and the lighting device **400**. The power supply controller **124** includes a battery detection circuit, a DC-DC converter, a switch circuit used for switching blocks to be energized, and so forth, and controls supply of electric power from the power supply section to the components of the image capturing apparatus **100**.

[0037] The first camera I/F **140** is an interface for communicably connecting between the system controller **120** and the external storage medium **200**. The second camera I/F **150** is an interface for communicably connecting between the system controller **120** and the lens controller **305**. The third camera I/F **170** is an interface for communicably connecting between the system controller **120**, and the strobe controller **402** and the video light controller **412**. The lens mount **160** is a mount portion on the side of the image capturing apparatus **100**, which is engaged with the camera mount **320** of the lens barrel **300** to thereby mechanically connect between the lens barrel **300** and the image capturing apparatus **100**.

[0038] The first camera connector **141** is connected to the media connector **203** of the external storage medium **200**. With this, the image capturing apparatus **100** and the external storage medium **200** are electrically connected to each other, whereby it is made possible to transmit a variety of control signals, image data, and so forth, and further, supply electric power from the image capturing apparatus **100** to the external storage medium **200**. The second camera connector **151** is connected to the lens connector **311**, and with this, the image capturing apparatus **100** and the lens barrel **300** are electrically connected to each other, whereby it is made possible to transmit a variety of control signals and state signals, data signals, and so forth, and supply electric power from the image capturing apparatus **100** to the lens barrel **300**. The third camera connector **171** is connected to the strobe connector **411**, and with this, the image capturing apparatus **100** and the lighting device **400** are electrically connected to each other, whereby it is possible to transmit a variety of control signals and state signals, data signals, and so forth, and supply electric power from the image capturing apparatus **100** to the lighting device **400**. Note that the second camera connector **151** and the third camera connector **171** can be configured to be capable of executing not only electrical communication, but also optical communication, audio communication, and so forth.

[0039] The external storage medium **200** is e.g. a memory card or a hard disk, and the storage section **201** is implemented e.g. by a semiconductor memory or a magnetic disk. The media connector **203** is electrically connected to the first camera connector **141** of the image capturing apparatus **100**. The media I/F **202** is an interface for communicably connecting between the external storage medium **200** and the system controller **120**.

[0040] The lens barrel **300** is an optical device which can be removably attached to the image capturing apparatus **100** and forms an optical image of an object on the image sensor **103**. The camera mount **320** is a mount portion on the side of the lens barrel **300**, which is engaged with the lens mount **160** of the image capturing apparatus **100** to thereby mechanically connect between the lens barrel **300** and the image capturing apparatus **100**. The lens connector **311** is provided inside the camera mount **320** and electrically connects between the lens barrel **300** and the image capturing apparatus **100**. The lens connector **311** can be configured to be capable of executing not only electrical communication, but also optical communication, audio communication, and so forth, in accordance with the configuration of the second camera connector **151**. The lens I/F **310** is an interface for communicably connecting between the lens controller **305** and the system controller **120**.

[0041] The lens controller **305** performs centralized control of the components of the lens barrel **300** according to a command from the system controller **120**. The photographic lens group **301** is comprised of a plurality of lenses, such as a focus lens, a zoom lens, and an image blur correction lens, and forms an image of incident light on an imaging surface of the image sensor **103**. The lens drive unit **304** adjusts the positions of the focus lens and the zoom lens in an optical axis direction according to an instruction from the lens controller **305**, and further, drives the image blur correction lens on a plane orthogonal to the optical axis. The diaphragm **302** adjusts the amount of light flux of an object, which is incident through the photographic lens group **301** toward the image sensor **103**. Note that the diaphragm **302** and the shutter **102** can be interlockingly controlled. The diaphragm drive unit **303** adjusts an opening amount of the diaphragm **302** according to an instruction from the lens controller **305**.

[0042] The lighting device **400** can be removably attached to the image capturing apparatus **100**, and irradiates an object with flash light according to a photographing environment, or illuminates (performs continuous irradiation of light on) the object. The strobe connector **411** is electrically connected to the third camera connector **171**. The strobe connector **411** can be configured to be capable of executing not only electrical communication, but also optical communication, audio communication, and so forth, in accordance with the configuration of the third camera connector **171**. The strobe I/F **410** is an interface for communicably connecting between the strobe controller **402** and the video light controller **412**, and the system controller **120**. The strobe light emission section **401** is e.g. a xenon tube. The strobe controller **402** controls light emission from the strobe light emission section **401** according to a command from the system controller **120**.

[0043] The video light **413** is implemented e.g. by an LED. The video light controller **412** controls light emission from the video light **413** according to a command from the system controller **120**. The light emission control of the video light **413** is performed by the PWM control, and the video light controller **412** notifies a frequency band which can be set in the PWM control to the system controller **120**. A variety of settings of the lighting device **400** can be made by an operation performed with respect to an operation section (not shown) included in the lighting device **400** and/or communication from the image capturing apparatus **100**.

[0044] Note that the processing blocks, such as the system controller **120**, the image processor **107**, the strobe controller **402**, the video light controller **412**, and the lens controller **305**, can be realized by the hardware, such as an ASIC and a programmable logic array (PLA). This is not limitative, but the processing blocks can be realized by a programmable processor, such as a CPU or MPU, which executes software (combination of the software and the hardware). That is, the same hardware can function as a plurality of functional blocks.

[0045] Next, image capturing operations in the image capturing system **1000** will be described. FIG. **2** is a flowchart of a shooting process performed by the image capturing system **1000**. Processing operations (steps) each denoted by S number in the flowchart are realized by the system controller **120** that performs centralized control of the image capturing system **1000** by executing a predetermined program. When a user turns on a power switch, not shown, of the lighting device **400** and the power switch **134** of the image capturing apparatus **100** to start the lighting device **400** and the image capturing apparatus **100**, the system controller **120** initializes the image capturing system **1000** and the initialization is completed, the system controller **120** starts the process.

[0046] When the initialization of the image capturing system **1000** is completed, in a step S**201**, the system controller **120** executes the live view (denoted as "LV" in FIG. **2**). In the live view, an image of an object is acquired, and the acquired image is displayed on the electronic viewfinder **105** or the rear display section **114**.

[0047] In a step S**202**, the system controller **120** determines whether the live view has been started by the operation of turning on the power switch **134**, or the live view temporarily interrupted in accordance with execution of predetermined processing has been restarted. Note that as an example of the restart, there are a case where moving image shooting has been terminated in a step S**209** and a case where predetermined processing, such as still image shooting, has been terminated in a step S**216**. If it is determined that the live view has been started, the system controller **120** executes a step S**203**, whereas if it is determined that the live view has been restarted, the system controller **120** executes a step S**204**.

[0048] In the step S**203**, the system controller **120** performs a flicker detection process. Details of the flicker detection process will be described hereinafter.

[0049] In the step S**204**, the system controller **120** determines a control mode of the video light **413** in the lighting device **400** and branches the process according to a result of the determination. The control mode of the video light **413** refers to a light emission mode of an auxiliary light for shooting, and the auxiliary light for shooting is irradiated on an object by lighting the video light **413**. In this step, whether the control mode is set to an automatic mode for automatically determining lighting-on/off of the video light **413**, a forcibly lighting mode for forcibly lighting the video light **413**, or a light-off mode for inhibiting lighting of the video light **413** is determined. The system controller **120** executes a step S**205** if it is determined that the control mode is set to the automatic mode, executes a lighting control process in a step S**206** if it is determined that the control mode is set to the forcibly lighting mode, and executes a step S**207** if it is determined that the control mode is set to the light-off mode.

[0050] In the step S**205**, the system controller **120** calculates a luminance value of the object from the image (frames) of the live view and determines whether or not it is necessary to turn on the video light **413**. In a case where the obtained luminance value is equal to or lower than a predetermined threshold value, the system controller **120** determines that it is necessary to turn on the video light **413** so as to irradiate the object with the auxiliary light for shooting. If it is determined that it is necessary to turn on the video light **413** (YES in S**205**), the system controller **120** executes the lighting control process in the step S**206**, whereas if it is determined that it is not necessary to turn on the video light **413** (NO in S**205**), the system controller **120** executes the step S**207**.

[0051] In the step S**206**, the system controller **120** instructs the video light controller **412** to turn on the video light **413**, and with this, the video light **413** is lighted. Note that details of the lighting control process in the **206** will be described hereinafter.

[0052] In the step S**207**, the system controller **120** determines whether or not an instruction for starting moving image shooting has been received. If it is determined that the moving image shooting start instruction has been received (YES in S**207**), the system controller **120** executes a step S**208**, whereas if it is determined that an instruction other than the moving image shooting start instruction has been received (NO in S**207**), the system controller **120** executes a step S**210**.

[0053] In the step S**208**, the system controller **120** performs moving image shooting (moving image recording).

[0054] In the step S**209**, the system controller **120** determines whether or not a moving image shooting termination instruction has been received. If it is determined that a moving image shooting termination instruction has not been received (NO in S**209**), the system controller **120** continues the processing in the step S**208**. If it is determined that a moving image shooting termination instruction has been received (YES in S**209**), the system controller **120** executes the step S**201**, and with this, the live view is restarted.

[0055] Description of the steps S**210** to S**212** is given assuming that three instructions other than the moving image shooting start instruction are exemplified, and further, an instruction other than these three instructions is a termination instruction provided by turning off the power switch **134** of the image capturing apparatus **100**.

[0056] In the step S**210**, the system controller **120** determines whether or not an instruction for starting still image shooting has been received. If it is determined that a still image shooting start instruction has been received (YES in S**210**), the system controller **120** executes a step S**214**, whereas if it is determined that an instruction other than the still image shooting start instruction has been received (NO in S**210**), the system controller **120** executes the step S**211**.

[0057] In the step S**211**, the system controller **120** determines whether or not an instruction for displaying a menu has been received. If it is determined that a menu display instruction has been received (YES in S**211**), the system controller **120** executes the step S**214**, whereas if it is determined that an instruction other than the menu display instruction has been received (NO in S**211**), the system controller **120** executes the step S**212**.

[0058] In the step S**212**, the system controller **120** determines whether or not an image reproduction display instruction has been received. The image to be reproduced may be a still image or a moving image. If it is determined that an image reproduction display instruction has received (YES in S**212**), the system controller **120** executes the step S**214**, whereas if it is determined that an instruction other than the image reproduction display instruction has been received (NO in S**212**), the system controller **120** executes a step S**213**.

[0059] In the step S**213**, the system controller **120** receives an instruction for turning off the power switch **134** and stops all functions of the image capturing system **1000**, followed by terminating the present process.

[0060] In the step S**214**, if the video light **413** is lighting, the system controller **120** turns off the video light **413**.

[0061] In a step S**215**, the system controller **120** interrupts the live view.

[0062] In the step S**216**, the system controller **120** executes predetermined processing according to the instruction received in the step S**210** to S**212**.

[0063] In a step S**217**, the system controller **120** determines whether or not a live view termination instruction has been received. If it is determined that a live view termination instruction has received (YES in S**217**), the system controller **120** waits until the next instruction is received. If it is determined that a live view termination instruction has not received (NO in S**217**), the system controller **120** executes the step S**201**, and with this, the live view is restarted.

[0064] Note that in the shooting process in FIG. **2**, in a case where it is determined in the step S**202** that the live view has been restarted, the flicker detection process in the step S**203** is not performed. This is not limitative, but the flicker detection process in the step S**203** can be performed regardless of whether or not the live view has been started or restarted without performing the determination processing in the step S**202**. In this case, in a case where the video light **413** has been turned on in the step S**206**, the video light **413** is turned off before execution of the flicker detection process.

[0065] Next, the flicker detection process in the step S**203** will be described. FIGS. **3**A and **3**B are diagrams useful in explaining a relationship between charge accumulation in the image sensor **103** and readout therefrom, and photometric values.

[0066] FIG. **3**A is a diagram showing charge accumulation control in the image sensor **103** with respect to flicker (having a frequency of 100 Hz) under a light which is lighted by a commercial power source of 50 Hz and changes in the photometric value. Note that flicker detection described hereafter is also used, as illustrated e.g. in FIG. **3**B, in a scene wherein a digital signage or the like subjected to lighting control at a predetermined frequency is present within a photographing angle of view, for detecting the lighting frequency of the digital signage.

[0067] To detect flicker, the image sensor **103** continuously performs charge accumulation and readout at a frame rate of 600 fps (=period of approximately 1.667 msec). Here, assuming that "n" is a natural number, charge accumulation at the n-th time is expressed as "accumulation n", readout of the accumulation n is expressed as "readout n", and a photometric value obtained from a result of the readout n is expressed as "AE (n)".

[0068] In FIG. **3**A, the charge accumulation operation is performed 12 times with the same exposure conditions, whereby the photometric values AE (1) to AE (12) are obtained. Note that charge accumulation in the image sensor **103** is performed within a limited time period, and hence the photometric values AE (1) to AE (12) are each represented by a median. An evaluation value SAD (m) used for flicker frequency determination is calculated from the photometric values AE (1) to AE (12) by using the following equation (1):

$$SAD(m)=\Sigma_{i=1}^{6}|AE(n)-AE(n+m)| \tag{1}$$

[0069] The evaluation value SAD (m) is one of indexes expressing similarity and is widely used e.g. in a pattern matching field. The value "m" is a value indicating what number-th photometric operation is performed ahead of a photometric operation performed for the photometric value AE (n) at the n-th time in the 12 photometric operations, to determine a photometric value of which similarity with the photometric value AE (n) is calculated. In FIG. **3**A, the evaluation value SAD (m) indicates the similarity with the photometric value obtained after the lapse of 1.667×m msec, and as the similarity is higher, the value of SAD (m) is smaller.

[0070] For example, in a case where the flicker period under a flicker environment, in which the frequency of the flicker is 100 Hz, is approximately 10 msec, and the charge accumulation time in the image sensor **103** is 1.667 msec, the relationship between these is expressed by 10/1.667≈6. This indicates that, as shown in FIG. **3**A, the same photometric value is obtained at every six periods regardless of the accumulation timing in the image sensor **103**. That is, as for the photometric value, a relationship expressed by AE (n)≈AE (n+6) holds. From this property, under the flicker environment in which the frequency of the flicker is 100 Hz, SAD (6)≈0 is determined.

[0071] To detect presence of flicker having a frequency of 100 Hz, SAD (3) indicating the similarity with a photometric value obtained after the lapse of 1.667×3=5 msec is further calculated. Under the flicker environment in which the frequency of the flicker is 100 Hz, a photometric value obtained at a timing shifted by 5 msec shows an opposite phase relation, and hence the value of SAD (3) is larger than the value of SAD (6). Therefore, in a case where SAD (3) is large, and SAD (6) is small, it is possible to determine that flicker having a frequency of 100 Hz is present.

[0072] Similarly, to detect flicker under a light which is lighted by a commercial power source of 60 Hz (the frequency of flicker is 120 Hz), SAD (5) and SAD (3) are only required to be calculated. FIG. **3**B is a diagram showing charge accumulation control in the image sensor **103** with respect to flicker under the light which is lighted by the commercial power source of 60 Hz and changes in the photometric value. The period of flicker having a frequency of 120 Hz is 8.333 msec, and hence AE (n)≈AE (n+5) holds, so that SAD (5)≈0 is determined.

[0073] Further, the flicker having a frequency of 120 Hz shows the opposite phase relation after the lapse of 4.16 msec, and hence it is preferable to determine the similarity with a photometric value obtained after the lapse of 4.16 msec. However, 4.16 msec is not a natural number multiple (integer multiple except 0 times) of the charge accumulation time of 1.667 msec. Then, as an example of a value relatively close to 4.16 msec, SAD (3) indicating the similarity with a photometric value obtained after the lapse of 5 msec is used. Also under the flicker environment in which the frequency of flicker is 120 Hz, SAD (3) indicates a photometric value close to an opposite-phase relation, and hence SAD (3) indicates a large value with respect to SAD (5).

[0074] As described above, in the flicker detection process, SAD (6), SAD (5), and SAD (3) are calculated, and presence/absence of flicker and the frequency in a case where flicker is present are detected based on the relationship between these values.

[0075] FIG. **4**A is a diagram showing an example of a map for determining whether or not flicker having a frequency of 100 Hz is detected. In a coordinate system having SAD (3) set to a horizontal axis and SAD (6) set to a vertical axis, respectively, a lower right area in which SAD (3) takes a large value relative to SAD (6) is set to an area A in which the frequency of flicker is determined as 100 Hz. It is possible to determine whether or not flicker having a frequency of 100 Hz is detected, based on whether or not a plot of SAD (3) and SAD (6), which are actually obtained, is included in the area A.

[0076] FIG. **4**B is a diagram showing an example of a map for determining whether or not flicker having a frequency of 120 Hz is detected. In a coordinate system having SAD (3) set to a horizontal axis and SAD (5) set to a vertical axis, respectively, a lower right area in which SAD (3) takes a large value relative to SAD (5) is set to an area B in which the frequency of flicker is determined as 100 Hz. It is possible to determine whether or not flicker having a frequency of 120 Hz is detected, based on whether or not a plot of SAD (3) and SAD (5), which are actually obtained, is included in the area B.

[0077] Note that a boundary line separating the area A and the other area is illustrated by way of example, and a bending point and inclination of the boundary line are not limited to those illustrated in FIG. **4**A, and the same is applied to FIG. **4**B.

[0078] FIG. **4**C is a table for finally determining presence/absence and the frequency of flicker, based on a result of the detection, obtained from FIGS. **4**A and **4**B. In a case where it is determined that the frequency of flicker is 100 Hz and at the same time is not 120 Hz, the frequency of flicker is finally determined as 100 Hz. Similarly, in a case where it is determined that the frequency of flicker is not 100 Hz and at the same time is 120 Hz, the frequency of flicker is finally determined as 120 Hz.

[0079] In a case where flicker is absent and stationary light (DC) is measured, the photometric value does not temporally change, and hence a relationship expressed by AE (1)≈AE (2)≈AE (3)≈ . . . ≈AE (12) is obtained. As a result, SAD (6)≈SAD (5)≈SAD (3)≈0 is obtained. Therefore, the photometric value of the stationary light is plotted in the vicinity of the origin in the respective coordinate systems shown in FIGS. **4**A and **4**B, and it is determined that the frequency of flicker is neither 100 Hz nor 120 Hz. As a result, in a case where it is determined that the frequency of flicker is neither 100 Hz nor 120 Hz, it is determined that the light source is the stationary light.

[0080] In general, it is normally difficult to consider that the flicker is determined to have a frequency of 100 Hz and also have a frequency of 120 Hz, but this determination result is sometimes obtained, for example, in a case where the photometric values AE (1) to AE (12) are acquired e.g. from a moving object or different objects due to a panning operation. In such a case, by regarding that an error has occurred, it is determined that the light source is the stationary light. With the above-described process, it is possible to determine presence/absence of flicker and whether the frequency in a case where flicker is present is 100 Hz or 120 Hz.

[0081] Next, the lighting control process for the video light **413** in the step S**206** will be described. Conventionally, exchange of the following information and data is performed between an image capturing apparatus and a lighting device. Specifically, from the image capturing apparatus to the lighting device, information on a charge accumulation time in the image sensor, an aperture value, an ISO sensitivity, a shooting mode, a lens focal distance, a light emission time in the lighting device, and an instruction for turning on/off the lighting device, is transmitted. Further, from the lighting device to the image capturing apparatus, information on a light emission mode, a guide number, and color temperature is transmitted.

[0082] In addition to these items of information, in the present embodiment, the image capturing apparatus **100** transmits an instruction of the frequency and the duty ratio (light emission amount) of the PWM control at a time when the video light **413** is lighted to the lighting device **400**. On the other hand, the lighting device **400** transmits the lighting state (the frequency and the duty ratio in the PWM control) of the video light **413** to the image capturing apparatus **100**.

[0083] FIG. **5** is a flowchart of the lighting control process for the video light **413**, which is performed in the step S**206**. In a step S**501**, the system controller **120** determines whether or not flicker has been detected in the step S**203**. If it is determined that flicker has not been detected (NO in S**501**), the system controller **120** executes a step S**502**, whereas if it is determined that flicker has been detected (YES in S**501**), the system controller **120** executes a step S**503**.

[0084] In the step S**502**, the system controller **120** calculates a frequency (or period) and a duty ratio in the PWM control for lighting the video light **413**, based on the charge accumulation time in the image sensor **103** at a time when the live view is executed and the luminance value of the object. Details of a method of determining the frequency (or period) and the duty ratio in the PWM control in the step S**502** will be described hereinafter.

[0085] In the step S**503**, the system controller **120** calculates a frequency (or period) and a duty ratio in the PWM control for lighting the video light **413**, based on the charge accumulation time in the image sensor **103** at a time when the live view is executed, the luminance value of the object, and the frequency of flicker detected in the step S**203**. Details of a method of determining the frequency (or period) and the duty ratio in the PWM control in the step S**503** will be described hereinafter.

[0086] When the processing in the step S**502** or S**503** is terminated, the system controller **120** executes a step S**504**. In the step S**504**, the system controller **120** transmits a result of the calculation in the step S**502** or S**503** (the frequency and the duty ratio in the PWM control for lighting the video light **413**) to the video light controller **412**.

[0087] In a step S**505**, the video light controller **412** lights the video light **413** by performing the PWM control at the frequency (period) and the duty ratio, which have been received from the system controller **120**. Note that the video light controller **412** notifies the lighting state of the video light **413** (the actual frequency and duty ratio in the PWM control) to the system controller **120** at fixed time intervals. This terminates the lighting control process for the video light **413** is terminated.

[0088] Next, the details of the methods used when determining the frequency and the duty ratio in the PWM control in the steps S**502** and S**503** will be described. Assuming that the frequency of the PWM control for the video light **413** is represented by "F", the charge accumulation time in the image sensor **103** is represented by "t", and the natural number is represented by "N", the frequency F of the PWM control in a case where flicker has not been detected is determined by $F=1/(t\times N)$. Further, the duty ratio for determining the light emission amount of the video light **413** is determined by using a known technique such that proper exposure is obtained with respect to the luminance value of the object. With this, it is possible to perform moving image shooting and live view, in which stripes generated due to lighting of the video light **413** are suppressed.

[0089] In a case where flicker has been detected in the step S**203**, in the present embodiment, the priority is given to adjusting the frequency of the PWM control for the video light **413** to the frequency of the detected flicker.

[0090] For example, if the frequency of the detected flicker is 100 Hz, the system controller **120** sets the frequency F of the PWM control for the video light **413** to 100 Hz. Then, the system controller **120** sets the charge accumulation time in the image sensor **103** to a natural number multiple of 1/100 second which is a period corresponding to the frequency of 100 Hz to suppress generation of stripes in a photographed image. Note that this method of suppressing generation of stripes is known (see e.g. Japanese Laid-Open Patent Publication (Kokai) No. 2009-213076), and hence detailed description thereof is omitted. The duty ratio for determining the light emission amount of the video light **413** is determined such that proper exposure is obtained with respect to the luminance value of the object, similarly to the case where flicker has not been detected.

[0091] However, in the case where flicker has been detected in the step S**203**, if it is impossible to adjust the frequency of the PWM control to the frequency of the detected flicker, the frequency of the PWM control is set similarly to the case where flicker has not been detected.

[0092] Note that in the case where flicker has been detected in the step S**203**, if the image capturing apparatus **100** is in a mode in which the shutter speed is not fixed, such as the aperture priority mode, it is possible to cope with the frequency adjustment as described above. On the other hand, in a case where the charge accumulation time in the image sensor **103** is fixed to a value which is not a natural number multiple of the period of flicker, the system controller **120** outputs a warning. The warning is provided e.g. by displaying the warning on the rear display section **114** or the electronic viewfinder **105**, outputting a warning sound in a case where a speaker is equipped, or displaying, in a case where the lighting device **400** includes a display section, the warning on this display section.

[0093] Further, the system controller **120** is required to determine the frequency F within a range of the frequency band notified from the video light controller **412**, which can be set in the PWM control. In a case where the video light controller **412** does not support the PWM control at the frequency F calculated based on the charge accumulation time calculated based on the fixed aperture value or the charge accumulation time fixed in advance, the system controller **120** outputs a warning. For example, as such a case, there can be mentioned a case where the video light controller **412** does not have the capability of generating a pulse wave of the frequency F. Note that the warning can be provided by using the above-mentioned methods.

[0094] In the above-described embodiment, the system controller **120** calculates the frequency F and the duty ratio in the PWM control for lighting the video light **413**. This is not limitative, but the video light controller **412** can calculate and determine the frequency F (or period) and the duty ratio in the PWM control for lighting the video light **413**. In this configuration, the system controller **120** transmits the charge accumulation time in the image sensor **103**, the luminance value of the object, a result of the flicker detection in the step S**203** (the information on presence/absence of flicker and the frequency of flicker if detected) to the video light controller **412**.

[0095] In a case where the video light controller **412** determines the frequency F, the video light controller **412** is not required to notify the frequency band which can be set in the PWM control to the system controller **120** but determines the frequency F within this frequency band. Then, the video light controller **412** transmits the set frequency F and information indicating whether or not the frequency F can be synchronized with the charge accumulation time in the image sensor **103** to the system controller **120**. In a case where the frequency F calculated based on the charge accumulation time in the image sensor **103** is outside the frequency band which can be set, the video light controller **412** further notifies a warning to the system controller **120**. The system controller **120** having received this warning can be configured to output a warning. Note that as the method of providing the warning, the above-described methods can be used.

[0096] The disclosure has been described heretofore based on the preferred embodiment thereof. However, the disclosure is not limited to the specific embodiment, but it is to be understood that the disclosure includes various forms within the scope of the gist of the disclosure. Further, the embodiment of the disclosure is described only by way of example, and it is possible to combine elements of the embodiment on an as-needed basis.

[0097] For example, although in the above-described embodiment, the configuration in which the lens barrel **300** and the lighting device **400** can be removably attached to the image capturing apparatus **100** has been described, the lens barrel **300** and the lighting device **400** can be integrally formed with the image capturing apparatus **100**. In this case, the system controller **120** also functions as the lens controller **305**, the strobe controller **402**, and the video light controller **412**. Further, although the configuration in which the lighting device **400** is physically connected to the image capturing apparatus **100** via the connector has been described, the lighting device **400** and the image capturing apparatus **100** can be controllably connected by wireless communication. Further, although as the image capturing apparatus **100** as a component of the image capturing system according to the disclosure, the mirrorless-type digital camera has been described, the image capturing apparatus **100** is not limited to this but can be a portable device, such as a digital video camera or a smartphone.

OTHER EMBODIMENTS

[0098] Embodiment(s) of the disclosure can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)™), a flash memory device, a memory card, and the like.

[0099] While the disclosure has been described with reference to exemplary embodiments, it is to be understood that the disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0100] This application claims the benefit of Japanese Patent Application No. 2024-025507 filed Feb. 22, 2024, which is hereby incorporated by reference herein in its entirety.

What is claimed is:

1. An image capturing system including an image capturing apparatus and a lighting device, which are communicably connected to each other,

wherein the image capturing apparatus comprises:

an image sensor; and

at least one memory configured to store instructions, and at least one processor in communication with the at least one memory and configured to execute the instructions to function as:

a first control unit configured to control a charge accumulation time in the image sensor,

wherein the lighting device comprises:

a light emission section; and

at least one memory configured to store instructions, and at least one processor in communication with the at least one memory and configured to execute the instructions to function as:

a second control unit configured to control light emission from the light emission section by performing PWM control,

wherein the first control unit determines a frequency in the PWM control, based on the charge accumulation time, and notifies the determined frequency to the second control unit, and

wherein the second control unit controls light emission from the light emission section at the frequency in the PWM control, which is notified from the first control unit.

2. The image capturing system according to claim **1**, wherein assuming that the frequency in the PWM control is represented by "F", the charge accumulation time is represented by "t", and a natural number is represented by "N", the frequency in the PWM control is determined by calculating $F=1/(t\times N)$.

3. The image capturing system according to claim **2**, wherein the instructions are executed to further function as a detection unit configured to detect flicker, and

wherein in a case where flicker is not detected by the detection unit, the frequency in the PWM control is determined by calculating $F=1/(t\times N)$, whereas in a case where flicker is detected by the detection unit, the frequency in the PWM control is determined to be the same frequency as a frequency of the detected flicker.

4. The image capturing system according to claim **3**, wherein in a case where flicker is detected by the detection unit, the charge accumulation time is set to a natural number multiple of a period of the detected flicker.

5. The image capturing system according to claim **3**, wherein in a case where flicker is detected by the detection unit, when it is impossible to set the frequency in the PWM control to the same frequency as the frequency of the detected flicker, the frequency in the PWM control is determined by calculating $F=1/(t\times N)$.

6. The image capturing system according to claim **1**, wherein the instructions are executed to further function as:

a detection unit configured to detect luminance information from an image acquired by the image sensor, and

a setting unit configured to set a duty ratio in the PWM control, based on the detected luminance information.

7. The image capturing system according to claim **1**, wherein the instructions are executed to further function as a warning unit configured to output a warning in a case where the second control unit does not support the frequency in the PWM control, which is determined based on the charge accumulation time.

8. An image capturing system including an image capturing apparatus and a lighting device, which are communicably connected to each other,

wherein the image capturing apparatus comprises:

an image sensor; and

at least one memory configured to store instructions, and at least one processor in communication with the at least one memory and configured to execute the instructions to function as:

a first control unit configured to control a charge accumulation time in the image sensor,

wherein the lighting device comprises:

a light emission section; and

at least one memory configured to store instructions, and at least one processor in communication with the at least one memory and configured to execute the instructions to function as:

a second control unit configured to control light emission from the light emission section by performing PWM control,

wherein the first control unit notifies the charge accumulation time to the second control unit, and

wherein the second control unit determines a frequency in the PWM control, based on the charge accumulation time, and controls light emission from the light emission section.

9. The image capturing system according to claim **8**, wherein assuming that the frequency in the PWM control is represented by "F", the charge accumulation time is represented by "t", and a natural number is represented by "N", the frequency in the PWM control is determined by calculating $F=1/(t\times N)$.

10. The image capturing system according to claim **9**, wherein the instructions are executed to further function as a detection unit configured to detect flicker, and

wherein in a case where flicker is not detected by the detection unit, the frequency in the PWM control is determined by calculating $F=1/(t\times N)$, whereas in a case where flicker is detected by the detection unit, the frequency in the PWM control is determined to be the same frequency as a frequency of the detected flicker.

11. The image capturing system according to claim **10**, wherein in a case where flicker is detected by the detection unit, the charge accumulation time is set to a natural number multiple of a period of the flicker.

12. The image capturing system according to claim **10**, wherein in a case where flicker is detected by the detection unit, when it is impossible to set the frequency in the PWM control to the same frequency as a frequency of the detected flicker, the frequency in the PWM control is determined by calculating $F=1/(t \times N)$.

13. The image capturing system according to claim **8**, wherein the instructions are executed to further function as:

a detection unit configured to detect luminance information from an image acquired by the image sensor, and

a setting unit configured to set a duty ratio in the PWM control based on the luminance information.

14. The image capturing system according to claim **8**, wherein the instructions are executed to further function as a warning unit configured to output a warning in a case where the second control unit does not support the frequency in the PWM control, which is determined based on the charge accumulation time.

15. An image capturing apparatus comprising:

an image sensor;

at least one memory configured to store instructions, and at least one processor in communication with the at least one memory and configured to execute the instructions to function as:

a communication unit configured to perform communication with a lighting device;

a determination unit configured to determine, assuming that the charge accumulation time in the image sensor is represented by "t", a natural number is represented by "N", and the frequency in the PWM control at a time when the lighting device controls light emission by performing the PWM control is represented by "F", the frequency in the PWM control by calculating $F=1/(t \times N)$, and

a notification unit configured to notify the frequency determined by the determination unit by using the communication unit to the lighting device.

16. An image capturing apparatus comprising:

an image sensor;

a light emission section configured to emit light toward an object; and

at least one memory configured to store instructions, and at least one processor in communication with the at least one memory and configured to execute the instructions to function as:

a control unit configured to control light emission from the light emission section by performing PWM control and control a charge accumulation time in the image sensor,

wherein assuming that the charge accumulation time is represented by "t", a natural number is represented by "N", and the frequency in the PWM control is represented by "F", the control unit determines the frequency in the PWM control by calculating $F=1/(t \times N)$.

17. A lighting device comprising:

a light emission section; and

at least one memory configured to store instructions; and at least one processor in communication with the at least one memory and configured to execute the instructions to function as:

a control unit configured to control the light emission section by performing PWM control;

a communication unit configured to perform communication with an image capturing apparatus; and

an acquisition unit configured to acquire a charge accumulation time set to an image sensor included in the image capturing apparatus, by using the communication unit,

wherein assuming that the charge accumulation time is represented by "t", a natural number is represented by "N", and the frequency in the PWM control is represented by "F", the control unit determines the frequency in the PWM control by calculating $F=1/(t \times N)$.

18. A method of controlling an image capturing system in which an image capturing apparatus and a lighting device are communicably connected to each other, comprising:

setting a charge accumulation time in an image sensor of the image capturing apparatus; and

determining, assuming that the charge accumulation time is represented by "t", a natural number is represented by "N", and a frequency in PWM control for causing a light emission section included in the lighting device to emit light is represented by "F", the frequency in the PWM control by calculating $F=1/(t \times N)$.

19. The method according to claim **18**, further comprising detecting flicker, and

wherein in a case where flicker is not detected, the frequency in the PWM control is determined by calculating $F=1/(t \times N)$, whereas in a case where flicker is detected, the frequency in the PWM control is determined to be the same frequency as a frequency of the flicker.

20. A non-transitory computer-readable storage medium storing a program for causing a computer to execute a method of controlling an image capturing system in which an image capturing apparatus and a lighting device are communicably connected to each other,

wherein the method comprises:

setting a charge accumulation time in an image sensor of the image capturing apparatus; and

determining, assuming that the charge accumulation time is represented by "t", a natural number is represented by "N", and a frequency in PWM control for causing a light emission section included in the lighting device to emit light is represented by "F", the frequency in the PWM control by calculating $F=1/(t \times N)$.

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